



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

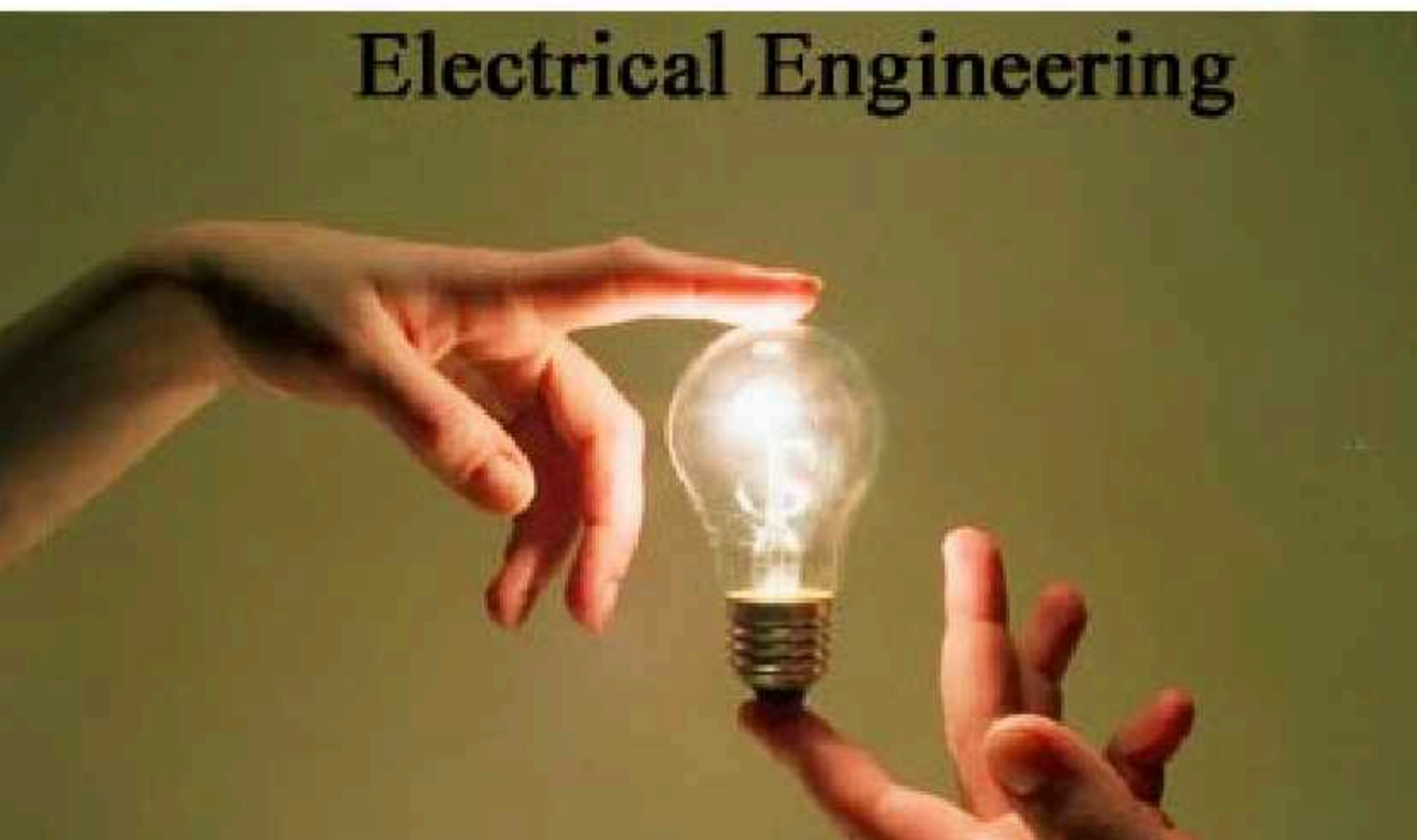
Accredited by NAAC & An ISO 9001: 2015 Certified Institution

T E C H N I C A L M A G A Z I N E T E C H I N N O V A T E

2022-2023

RIT - EEE

Electrical Engineering



Chief Editor

Mr. D. Karthik Prabhu, Assistant Professor.

Student Coordinator(s)

Ms. B. Jeyapadma , III year - EEE

Ms. G. Dhana Bharathi Lakshmi, III year - EEE

Vision, Mission, PEOs & PSOs

VISION

- To create competent electrical and electronics engineers with professional and research skills, and a sense of ethics to meet the demands of multidisciplinary challenges and a technologically advancing society.

PEOs

- Graduates will pursue their careers in diversified sectors of the industry, higher education in their fields of interest and/or emerge as entrepreneurs.
- Graduates will design, manufacture, and/or maintain systems in the field of electrical and electronics engineering and allied engineering industries.
- Graduates will work effectively as individuals and as team members in multidisciplinary projects with leadership qualities, social and ethical responsibilities.

MISSION

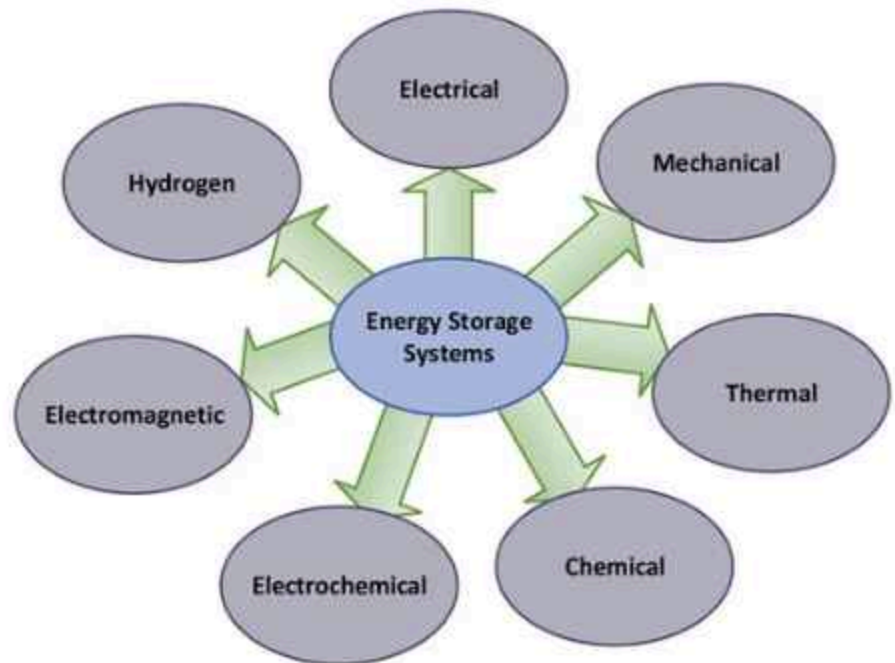
- Empowering the students with up-to-date knowledge and technological skills in Electrical and Electronics Engineering thrust areas.
- Providing a rigorous practical training through industrial collaboration to produce Electrical and Electronics Engineers with good design experience.
- Adopting effective teaching learning methodologies and value added courses to create Electrical and Electronics Engineers with a strong fundamental knowledge.
- Enabling the students to be life-long learners to meet intellectual, ethical and career challenges.

PSOs

- PSO1: Analyse the performance of electrical and electronic systems using modern software packages.
- PSO2: Specify, design, prototype and test electronic systems and embedded systems in various engineering applications.
- PSO3: Design and develop the hardware and software required for industrial automation systems.

Energy Storage

Energy storage refers to the capture and retention of energy for later use. It plays a crucial role in the integration of renewable energy sources, such as solar and wind, into the power grid, as these sources are intermittent and may not always align with the demand for electricity. Energy storage technologies enable the storing of excess energy when it's abundant and releasing it when needed, helping to balance the supply and demand of electricity.



There are various types of energy storage technologies, each with its own advantages and disadvantages. Some common forms of energy storage include:

1. Battery Storage:

- Lithium-ion batteries are widely used for applications ranging from small electronics to electric vehicles and grid-scale energy storage.
- Other battery technologies include lead-acid, sodium-ion, and flow batteries.

2. Pumped Hydro Storage:

- In this method, excess electricity is used to pump water from a lower reservoir to an upper reservoir during periods of low demand. When electricity is needed, the stored water is released to generate power by flowing downhill through turbines.

3. Flywheel Energy Storage:

- Rotating flywheels store energy in the form of kinetic energy. When energy is needed, the flywheel's rotation is converted back into electricity.

4. Compressed Air Energy Storage (CAES):

- Excess electricity is used to compress air and store it in underground caverns. When electricity is needed, the compressed air is released, heated, and expanded to drive turbines that generate power.

5. Thermal Energy Storage:

- This involves storing heat or cold for later use. Common methods include using molten salt for solar thermal storage or storing ice for cooling purposes.



WIND FARM

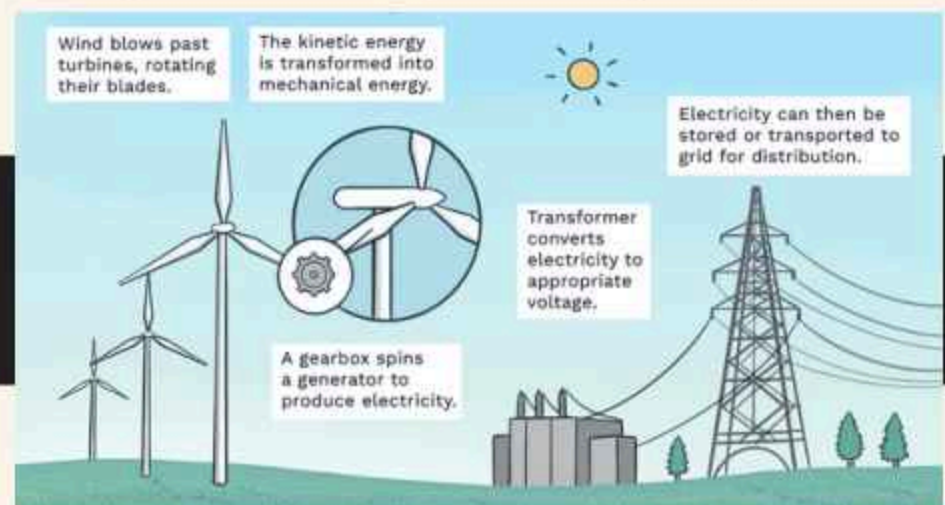
Installed Capacity Growth: India has witnessed significant growth in its wind power capacity over the years. By the end of 2021, India had an installed wind power capacity of over 38 gigawatts (GW), making it one of the largest wind power markets globally.

Geographical Distribution: Wind farms are spread across various states in India, with some states having higher wind power potential than others. States like Tamil Nadu, Maharashtra, Gujarat, Karnataka, and Rajasthan are among the leaders in terms of installed wind power capacity.

Technological Advancements: There have been advancements in wind turbine technology and project management practices in India. Modern turbines are more efficient and reliable, leading to improved performance of wind farms across the country.

Hybrid Projects: There's a growing trend towards hybrid renewable energy projects in India, which combine wind power with solar power or other sources. These projects help optimize land use, grid integration, and resource utilization.

Grid Integration: Grid integration of wind power has been a focus area for India, as the variability of wind resources presents challenges for grid stability. Efforts are underway to enhance grid infrastructure and develop forecasting and scheduling mechanisms to manage the variability of wind power.



Latest update

In recent news, offshore wind farms continue to expand globally, with significant developments in various regions. The United States announced plans to accelerate offshore wind energy projects, aiming to deploy 30 gigawatts by 2030. In Europe, the Netherlands inaugurated its largest offshore wind farm, Borssele 1 & 2, boosting its renewable energy capacity. Additionally, advancements in turbine technology, including larger rotor diameters and floating platforms, are enhancing offshore wind farm efficiency and viability in deeper waters. These developments underscore the growing importance of offshore wind energy in the transition to a sustainable and low-carbon future worldwide.

Offshore Wind Farms

Location: Offshore wind farms are typically located in coastal areas where there are strong and consistent wind resources. These locations often benefit from higher wind speeds compared to onshore sites, resulting in increased electricity generation potential.

Advantages:

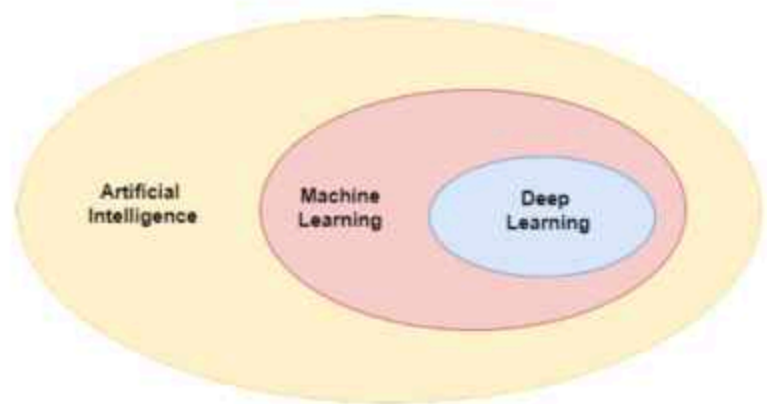
Stronger Winds: Offshore wind farms have access to stronger and more consistent wind speeds compared to onshore locations, leading to higher energy yields.

Land Use: Offshore wind farms do not compete for land resources, making them suitable for densely populated regions with limited land availability.

THE POWER OF DL AND ML

Deep learning (DL) and machine learning (ML) are two interconnected fields within artificial intelligence (AI) that have seen rapid advancements and widespread applications in various industries

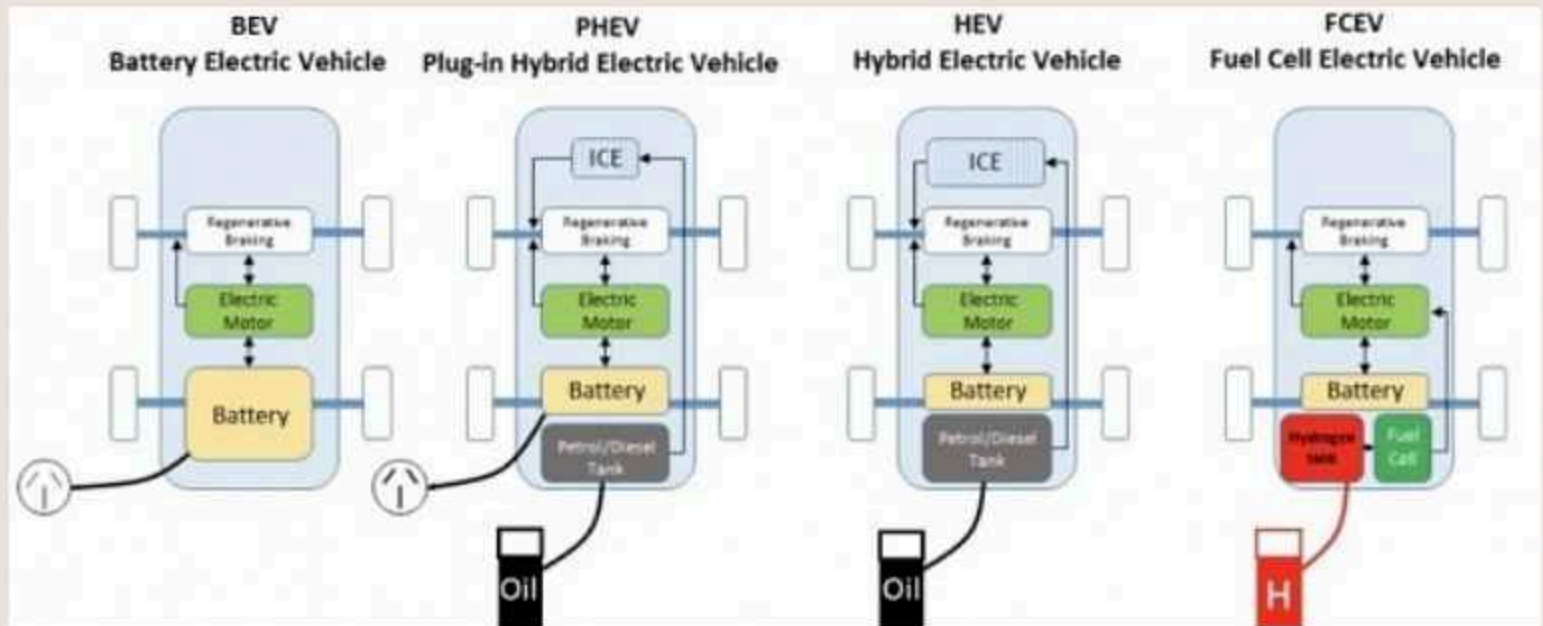
Machine Learning (ML): Machine learning is a subset of artificial intelligence that focuses on the development of algorithms and statistical models that enable computers to perform tasks without explicit programming. ML algorithms learn from data, identify patterns, and make decisions or predictions based on that data. Common techniques in ML include supervised learning, unsupervised learning, and reinforcement learning.



Deep Learning (DL): Deep learning is a subfield of ML that uses neural networks with multiple layers (hence the term "deep") to learn representations of data in a hierarchical manner. DL algorithms are particularly effective for tasks such as image and speech recognition, natural language processing, and reinforcement learning. Deep learning has achieved remarkable success in recent years, driven by advances in computational power, big data, and algorithmic improvements.

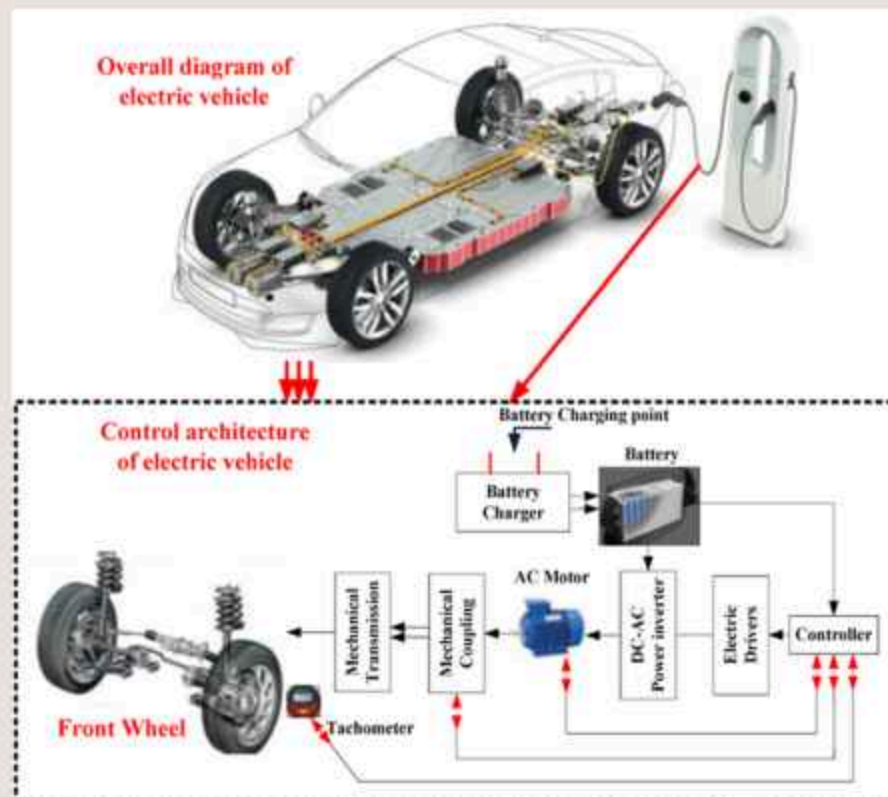
Data Science	AI Artificial Intelligence	• Reactive Machines • Limited Memory • Theory of Mind • Self-awareness
	ML Machine Learning	• Supervised Learning • Unsupervised Learning • Reinforcement Learning
	DL Deep Learning	• Convolutional Neural Network (CNN) • Recurrent Neural Network (RNN) • Generative Adversarial Network (GAN) • Deep Belief Network (DBN)

E-VEHICLE DIFFERENT ARCHITECTURE



Electric vehicles (EVs) encompass a variety of architectures, each offering unique advantages and design considerations. Battery Electric Vehicles (BEVs) rely solely on batteries for power, providing zero-emission transportation and typically offering longer ranges. Plug-in Hybrid Electric Vehicles (PHEVs) combine internal combustion engines with electric propulsion, offering flexibility by allowing drivers to switch between electric and gasoline power. Hybrid Electric Vehicles (HEVs) use both an internal combustion engine and an electric motor, with the latter primarily assisting the engine for improved fuel efficiency. Fuel Cell Electric Vehicles (FCEVs) utilize hydrogen fuel cells to generate electricity, emitting only water vapor as a byproduct and offering fast refueling times. E-axle architecture integrates electric motor, power electronics, and transmission into a single unit, simplifying vehicle design and improving efficiency. Each architecture presents unique challenges and opportunities, influencing factors such as range, charging infrastructure requirements, and overall performance, driving innovation in the rapidly evolving landscape of electric mobility.

Electric vehicles (EVs) employ diverse architectures, including Battery Electric Vehicles (BEVs) with exclusive battery power, Plug-in Hybrid Electric Vehicles (PHEVs) merging electric and gasoline power, and Hybrid Electric Vehicles (HEVs) blending internal combustion and electric propulsion. Fuel Cell Electric Vehicles (FCEVs) utilize hydrogen fuel cells, advancing emission-free transportation.



SMART AGRICULTURE SYSTEM USING IOT

What is IoT?

The Internet of Things (IoT) describes the network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.



Among the technologies available for present-day farmers are:

Sensors: soil, water, light, humidity, temperature management

Software: specialized software solutions that target specific farm types or applications agnostic IoT platforms

Connectivity: cellular, LoRa

Location: GPS, Satellite

Robotics: Autonomous tractors, processing facilities

Data analytics: standalone analytics solutions, data pipelines for downstream solutions



Smart farming refers to managing farms using modern Information and communication technologies to increase the quantity and quality of products while optimizing the human labor required.

Power Electronics Converters for E-Vehicles

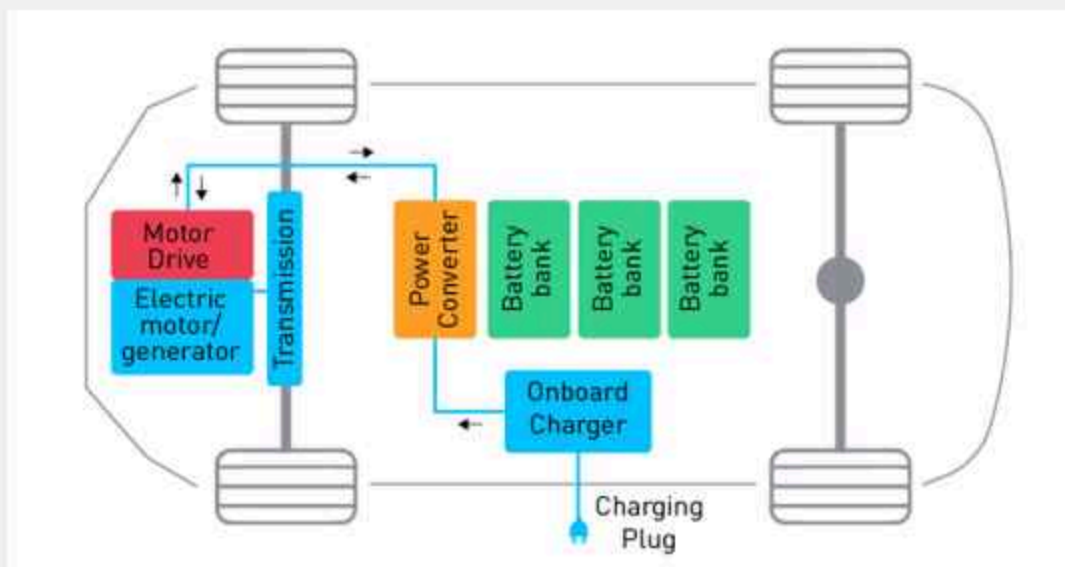
DC-DC Converters: DC-DC converters are used to regulate the voltage between different components in an EV system. They step down the high-voltage DC output from the battery to supply lower voltage levels required by auxiliary systems such as lighting, heating, and cooling.

AC-DC Converters: AC-DC converters, also known as rectifiers, are used to convert AC power from the grid or onboard generators (e.g., fuel cells) into DC power to charge the EV battery. They are typically integrated into the onboard charger of the EV and support different charging standards (e.g., AC Level 1, AC Level 2).

DC-AC Inverters: DC-AC inverters are used to convert the DC power from the battery into AC power to drive the electric motor. They control the frequency, amplitude, and phase of the AC output to regulate the speed and torque of the motor, enabling smooth acceleration and efficient operation of the EV.

Bidirectional DC-DC Converters: Bidirectional DC-DC converters enable energy flow in both directions between the battery and other components, such as the motor or the grid. They support regenerative braking, where kinetic energy from braking is converted back into electrical energy and stored in the battery.

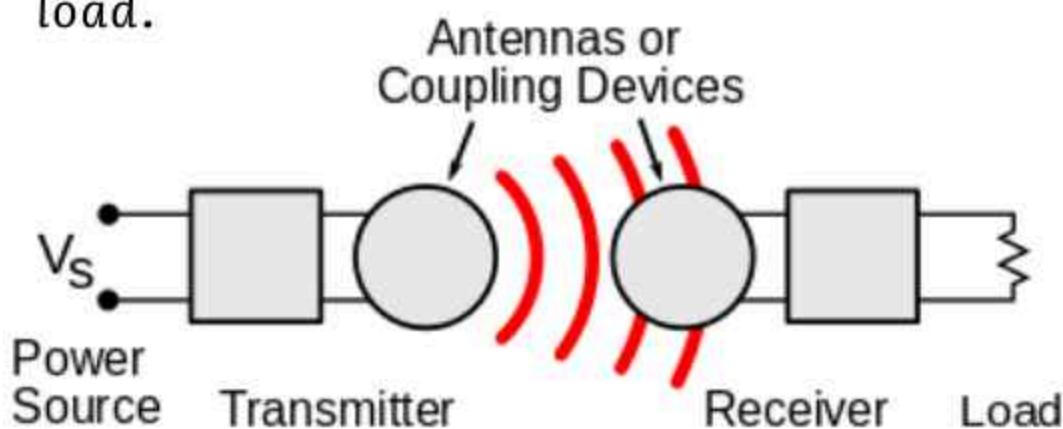
Voltage Source Inverters (VSIs): VSIs are a type of power electronic converter used in EV traction systems to control the output voltage and frequency of the motor. They convert the DC voltage from the battery into three-phase AC voltage to drive the motor, offering precise control over motor speed and torque. Overall, power electronics converters are essential for the efficient and reliable operation of electric vehicles, enabling energy conversion, management, and control to meet the demands of modern transportation. Advanced control strategies are implemented via power electronics to regulate the charge rate and protect the battery from potential harm.



Wireless Power Transmission

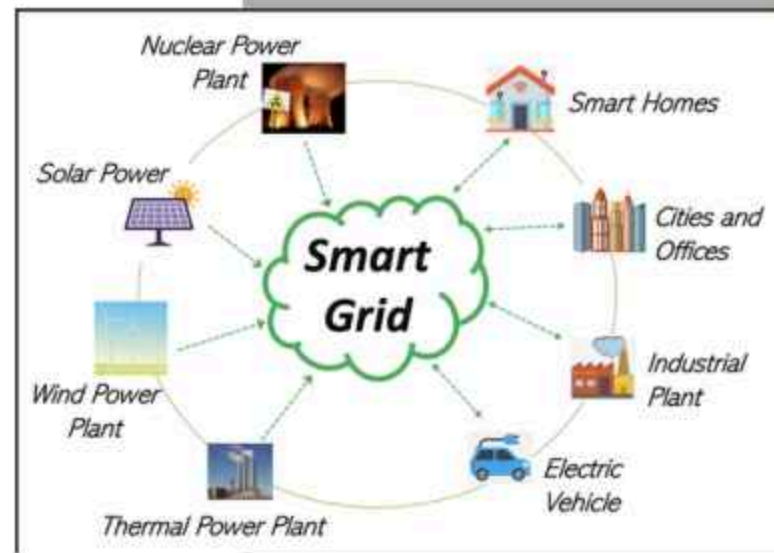
Wireless power transfer (WPT), wireless power transmission, wireless energy transmission (WET), or electromagnetic power transfer is the transmission of electrical energy without wires as a physical link. In a wireless power transmission system, an electrically powered transmitter device generates a time-varying electromagnetic field that transmits power across space to a receiver device; the receiver device extracts power from the field and supplies it to an electrical load.

The technology of wireless power transmission can eliminate the use of the wires and batteries, thereby increasing the mobility, convenience, and safety of an electronic device for all users. Wireless power transfer is useful to power electrical devices where interconnecting wires are inconvenient, hazardous, or are not possible. Wireless power techniques mainly fall into two categories: near field and far-field.



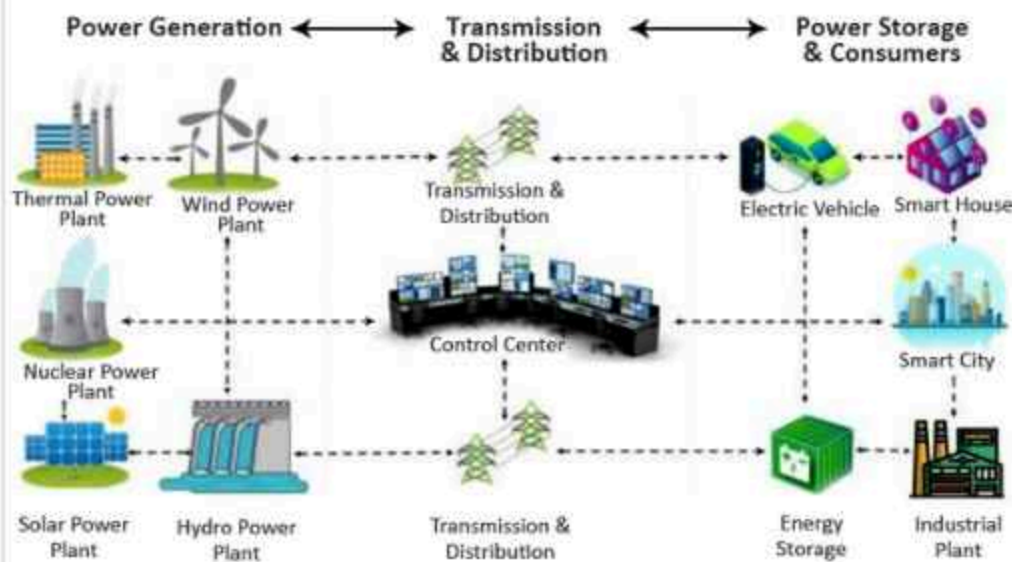
SMART GRID

A smart grid is an advanced electricity network that utilizes digital technology, two-way communication, and real-time data to optimize the generation, distribution, and consumption of electricity. Unlike traditional grids, smart grids integrate renewable energy sources, energy storage systems, and demand-side management techniques to enhance efficiency, reliability, and sustainability.



Key components of a smart grid include advanced metering infrastructure (AMI), which enables utilities to remotely monitor and control electricity consumption, and distribution automation systems, which detect and respond to faults in the grid automatically. Additionally, smart grids utilize sensors, smart appliances, and grid-connected devices to gather data on energy usage patterns and facilitate demand response programs, where consumers adjust their electricity consumption in response to price signals or grid conditions. Benefits of smart grids include improved grid resilience, reduced energy losses, and enhanced integration of renewable energy resources.

Smart Grid : The Electrical Grid of the Future



- Grid Modernization Initiatives:** Many countries are investing in grid modernization initiatives to upgrade aging infrastructure and integrate advanced technologies. These initiatives involve the deployment of smart meters, sensors, and automation systems to improve grid reliability, resilience, and efficiency.
- Renewable Energy Integration:** Smart grids play a crucial role in integrating renewable energy sources such as solar and wind power into the electricity grid. Advanced forecasting, grid management, and storage technologies help mitigate the variability of renewable energy generation and optimize its integration into the grid.
- Microgrids and Distributed Energy Resources (DERs):** Microgrids, which are localized grids capable of operating independently or in conjunction with the main grid, are gaining traction as a solution for enhancing grid resilience and supporting decentralized energy generation. Smart grid technologies enable the integration and management of DERs, including rooftop solar panels, energy storage systems, and electric vehicle chargers, within microgrid frameworks.

